VALIDATION REPORT
OPERATIONAL QUALIFICATION
CURVE MODULE

JR Validated Environment — Curve Module v1.1

Field	Value
Document Number	JR-VR-CURVE-001
Title	Validation Report — Operational Qualification, Curve Module
Validation Plan	JR-VP-CURVE-001 v1.1
System	JR Validated Environment — Curve Module
Module Version	1.1
Document Version	1.1
Execution Date	2026-06-10
OQ Result	PASS — 36 / 36 tests passed, 0 failures, 0 deviations
Author	Joep Rous
Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-03-26	Joep Rous	Initial OQ execution report. All 28 test cases passed. No deviations.
1.1	2026-06-10	Joep Rous	Module v1.1 OQ execution: robustness fixes and functionality extensions (X transforms, width_at_y, tangency classification, query plot markers, uniform-grid resampling). Suite expanded to 36 test cases per JR-VP-CURVE-001 v1.1. 36 / 36 tests passed.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-06-10
Reviewer	[Name]		
Approver (QA)	[Name]		

1. Purpose

This document reports the execution and outcome of the Operational Qualification (OQ) test suite for the JR Validated Environment Curve Module, version 1.1. It provides objective evidence that jrc_curve_properties performs as specified in the OQ Validation Plan JR-VP-CURVE-001 v1.1 under the defined test conditions.

This report records the test environment, execution results for all 36 test cases, and the formal OQ conclusion. Results are derived directly from the evidence file generated by admin_curve_oq.

2. Scope

2.1 In Scope

- The Python script repos/curve/jrc_curve_properties.py
- Config-file parsing and pre-validation (unknown sections/keys, non-numeric values, missing required keys)
- Global statistics: max Y, min Y, max X, min X, AUC (trapezoidal), hysteresis
- Per-phase statistics with ascending/descending search and after_phase restriction
- Slope computation: overall linear regression, secant, numerical derivative at x
- Query interpolation: Y at x, X at y (first/last/all, with crossing/tangency classification), Y at relative x, and width at y (peak width / FWHM)
- Data transforms: x_scale, x_offset, y_scale, and y_offset_x (applied in that order)
- Transition detection: inflection points (d2y sign change) and yield point (slope threshold)
- Uniform-grid resampling for derivative calculations on non-uniformly sampled data ([smoothing] resample = yes)
- Output file creation: results .txt, optional plot PDF, optional debug d2y CSV
- Help mode when invoked with no arguments
- Numeric correctness: AUC, overall slope, and Y-at-x verified against independently computed reference values

2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by JR-IQ-001)
- Performance qualification (PQ) with real production data
- Validation of Python interpreter and third-party packages (scipy, numpy, matplotlib)
- Statistical validation of the Savitzky-Golay smoothing algorithm — algorithm reference cited in JR-VP-CURVE-001 Section 3
- Plot aesthetics and PDF rendering quality beyond file creation

3. Reference Documents

Document ID	Title	Location
JR-VP-CURVE-001 v1.1	Validation Plan — OQ, Curve Module	repos/curve/docs/curve_validation_plan.pdf
JR-VP-002 v1.0	Validation Plan — OQ, Community Script Suite v1.1.0	docs/oq_validation_plan.docx
JR-VP-001 v1.0	Validation Plan — Installation Qualification	docs/
JR-IQ-001	IQ Execution Evidence	docs/IQ_validation_20260311_205146.txt

OQ Evidence	Curve OQ Execution Evidence (pytest output + header)	curve_oq_execution_20260610T122336.txt

4. Test Environment

Component	Details
Execution date / time	2026-06-10T12:23:36
Host	Joeps-MacBook-Air.local
Operating System	26.5
Python Version	3.11.9
pytest Version	pytest 8.3.4
Test runner	repos/curve/admin_curve_oq
Test directory	repos/curve/oq/
Test data	repos/curve/oq/data/ (4 CSV fixtures, 19 config fixtures)
Execution command	bash repos/curve/admin_curve_oq
OQ evidence file	curve_oq_execution_20260610T122336.txt

5. Test Results

All 36 test cases were executed on 2026-06-10. Results are derived from the OQ evidence file and reproduced below. The evidence file contains the complete pytest output including individual test IDs, pass/fail status, and timing.

Test Class	Tests Executed	Result
TestCurveValidation (V)	7	7/7 PASS
TestCurveGlobal (G)	5	5/5 PASS
TestCurvePhase (P)	3	3/3 PASS
TestCurveSlope (S)	3	3/3 PASS
TestCurveQuery (Q)	6	6/6 PASS
TestCurveTransform (T)	3	3/3 PASS
TestCurveTransitions (I)	2	2/2 PASS
TestCurveOutputFiles (O)	3	3/3 PASS
TestCurveResample (R)	1	1/1 PASS
TestCurveNumeric (N)	3	3/3 PASS
TOTAL	36	36/36 PASS

5.1 TestCurveValidation — 7/7 PASS

TC ID	Description	Expected	Result
TC-CURVE-V-001	No arguments — help mode	exit 0, 'Usage' or 'USAGE' in output	**PASS**
TC-CURVE-V-002	Config file not found	exit non-zero	**PASS**
TC-CURVE-V-003	Data file not found	exit non-zero	**PASS**
TC-CURVE-V-004	Unknown section in config	exit non-zero, 'Unknown section' in output	**PASS**
TC-CURVE-V-005	Unknown key in [global]	exit non-zero, 'Unknown key' in output	**PASS**
TC-CURVE-V-006	Non-numeric value in [transform]	exit non-zero	**PASS**
TC-CURVE-V-007	[debug] d2y=yes without	exit non-zero	**PASS**

	d2y.phase		

5.2 TestCurveGlobal — 5/5 PASS

TC ID	Description	Expected	Result
TC-CURVE-G-001	max_y on linear data	exit 0, 'max Y' in output	**PASS**
TC-CURVE-G-002	min_y on linear data	'min Y' in output	**PASS**
TC-CURVE-G-003	max_x and min_x on linear data	'max X' and 'min X' in output	**PASS**
TC-CURVE-G-004	AUC on linear data (y=2x, x=0..20)	'AUC' in output, '400' in output	**PASS**
TC-CURVE-G-005	Hysteresis between loading and unloading phases	exit 0, 'Hysteresis' in output	**PASS**

5.3 TestCurvePhase — 3/3 PASS

TC ID	Description	Expected	Result
TC-CURVE-P-001	Per-phase max_y on loading phase	'max Y [loading]' in output	**PASS**
TC-CURVE-P-002	Per-phase AUC on loading phase	'AUC [loading]' in output	**PASS**
TC-CURVE-P-003	Multiple phases in one run	'loading' and 'unloading' both in output	**PASS**

5.4 TestCurveSlope — 3/3 PASS

TC ID	Description	Expected	Result
TC-CURVE-S-001	Overall slope on linear data (y=2x)	'slope overall' in output, '2' in output	**PASS**
TC-CURVE-S-002	Secant slope between two x points	'slope secant' in output	**PASS**
TC-CURVE-S-003	Numerical derivative slope at x	'slope at x=' in output	**PASS**

5.5 TestCurveQuery — 6/6 PASS

TC ID	Description	Expected	Result
TC-CURVE-Q-001	Y at x interpolation on linear data	'Y at x=' in output, '10' in output (y at x=5)	**PASS**
TC-CURVE-Q-002	X at y interpolation	'X at y=' in output	**PASS**
TC-CURVE-Q-003	Y at relative x interpolation	'Y at ' in output	**PASS**
TC-CURVE-Q-004	Width at y on triangle data	width = 5/9 ± 0.001 (crossings at x=5 and x=50/9)	**PASS**
TC-CURVE-Q-005	Tangency classification at triangle apex	'tangency' annotation in output for y=100 touch point	**PASS**
TC-CURVE-Q-006	Query .show markers drawn on plot	exit 0, width_plot.pdf created	**PASS**

5.6 TestCurveTransform — 3/3 PASS

TC ID	Description	Expected	Result
TC-CURVE-T-001	y_scale transform applied	exit 0, 'Y scale' in output	**PASS**
TC-CURVE-T-002	y_offset_x transform applied	'Y offset' in output	**PASS**
TC-CURVE-T-003	x_scale and x_offset	'X scale' and 'X offset' in output; slope	**PASS**

	transforms applied	= 1.000 ± 0.001	

5.7 TestCurveTransitions — 2/2 PASS

TC ID	Description	Expected	Result
TC-CURVE-I-001	Inflection points on sine data	'inflection' in output	**PASS**
TC-CURVE-I-002	Yield point detected on sine data	'yield point' in output	**PASS**

5.8 TestCurveOutputFiles — 3/3 PASS

TC ID	Description	Expected	Result
TC-CURVE-O-001	Results .txt file created	exit 0, _results.txt file exists	**PASS**
TC-CURVE-O-002	Plot PDF created when plot=yes	exit 0, .pdf file exists	**PASS**
TC-CURVE-O-003	Debug d2y CSV created when d2y=yes	exit 0, _debug_d2y_<phase>.csv file exists	**PASS**

5.9 TestCurveResample — 1/1 PASS

TC ID	Description	Expected	Result
TC-CURVE-R-001	Uniform-grid resampling on non-uniformly sampled sine data	inflection at x = π ± 0.1; no non-uniform spacing warning	**PASS**

5.10 TestCurveNumeric — 3/3 PASS

These tests verify that computed values match independently derived reference values within specified tolerances. Dataset: linear.csv (y=2x, x=0..20).

TC ID	Description	Expected	Result
TC-CURVE-N-001	AUC for linear.csv (y=2x, x=0..20)	400.0 ± 0.5 (trapezoid rule, exact linear)	**PASS**
TC-CURVE-N-002	Overall OLS slope for linear.csv (y=2x)	2.000 ± 0.001 (OLS on y=2x)	**PASS**
TC-CURVE-N-003	Y at x=5.0 for linear.csv (y=2x)	10.0 ± 0.01 (exact point in data)	**PASS**

6. Deviations

No deviations were identified during OQ execution. All 36 test cases passed on execution.

7. OQ Conclusion

The Operational Qualification of the JR Validated Environment Curve Module v1.1 is complete.

Result: PASS

All 36 automated test cases specified in JR-VP-CURVE-001 v1.1 were executed on 2026-06-10 and 36 passed. Results are derived from the OQ evidence file listed below.

- jrc_curve_properties — XY Curve Analysis (Config-file driven)

Evidence of test execution is archived at: curve_oq_execution_20260610T122336.txt

The Curve module is released as version 1.1 of repos/curve/. Any subsequent change to the script, wrapper, or help file within repos/curve/ will require a change assessment and, if the change affects functionality covered by this OQ, re-execution of the affected test cases.